

Instrumentation Engineering: A Vast Field That's Growing

Today the environment, be it in any field, is fast-changing. Had it not been for Instrumentation engineers, we might have remained in the 'Dark Age' of electronics. Because without the instruments we have today to verify things, we would have been stuck with imperfect communication technology. Let us find out the scope for a career in Instrumentation in India



(Image courtesy: www.karunya.edu)

ABHISHEK MUTHA

The instruments we see today were either non-existent or worked very differently ten to fifteen years ago. It would not be wrong to say that the field of instrumentation is a superset of the fast-changing fields of electronics, information technology, measurement and electronic communication. Basically, it is a multi-disciplinary stream.

Instrumentation engineering is loosely defined because the required tasks are very domain-dependent. For instance, a biomedical instrumentation expert has very different concerns

from a rocket instrumentation expert, though both might have some common concerns such as selection of sensors based on various parameters.

"Instrumentation has been the fundamental engine for catalysing R&D in India. Segments like communication, aerospace and defense, industrial, computers and semiconductors, education and academia, and general-purpose electronics also use instrumentation to validate their devices, designs and deployment," informs Jyotsna Kukreti Dobriyal, HR operations, Agilent Technologies India.

"New concepts of instrumentation-aware systems have opened this do-

main even further. Engineers at any level, whether it is design, validation or software development, need proper knowledge of test methods. This domain is growing in India," she adds.

Hemant Kamat, chief technology officer, Shalaka Technologies, too believes that instrumentation is a very important aspect of manufacturing industry. "Instrumentation is applied in R&D, process, manufacturing and quality departments of manufacturing organisations. It is also important in building management systems—a market that is growing significantly in India," he shares.

Scope, roles and responsibilities

Unlike services vertical, instrumentation industry is not manpower-intensive. However, it requires talented individuals that have specific skills in measurement science and engineering, believes Jyotsna. "Those who are sound in fundamentals and have specific knowledge of measurement systems will find multiple career options within this industry," she says.

With the increasing demand for electrical power and developments in audio and video communication systems and the automation industry, electrical/instrumentation engineers have become vital to the society today.

"Industries such as textile, pharmaceutical, paper, metallurgy, automotive, marine, locomotive and